



GENERAL CONSTRUCTION PERMIT

Oil and Gas Industry

Natural Gas Fired

Reciprocating Internal Combustion Engines (RICE)

Permit Number GP02 Issuance 4

Final Approval

July 23, 2025

Bradley Eades, PE

Date Issued

Oil & Gas Permit Section Supervisor

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I. General Permit Applicability

I.A. This General Permit can be used only for natural gas fired reciprocating internal combustion engines located at an oil and gas stationary source that meets the following requirements:

I.A.1. Are a true minor or synthetic minor source for Operating Permits (OP) or New Source Review (NSR) program applicability, except as provided for in I.B, I.C, and I.D, below.

I.A.2. Are a true minor or synthetic minor source under the provisions of 40 CFR Part 63, National Emission Standards for Hazardous Air Pollutants for Source Categories.

I.A.3. Meet all of the conditions of Section II.A. of this General Permit.

I.B. Stationary Sources that became subject to Title V Operating Permit requirements or became classified as an existing major stationary source for which a complete GP02 registration request was received by the Division prior to the effective date (January 27, 2020) on which 8-hour Ozone Control Area was re-classified from moderate to serious may continue to operate under this General Permit. 8-hour



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Ozone Control Area is as defined in Regulation Number 7, Part A, Section II.A.1. (Adopted 12/14/2023, Effective 02/14/2024).

- I.C. Stationary Sources that: a) became subject to Title V Operating Permit requirements or became classified as an existing major stationary source due to the reclassification of the 8-hour Ozone Control Area from serious to severe, and b) for which a complete GP02 registration request was received by the Division prior to the effective date of GP02, Issuance 3 (November 3, 2022) may continue to operate under this General Permit. 8-hour Ozone Control Area is as defined in Regulation Number 7, Part A, Section II.A.1. (Adopted 12/14/2023, Effective 02/14/2024).
- I.D. Stationary Sources that: a) became subject to Title V Operating Permit requirements or became classified as an existing major stationary source for which a GP02 registration request was deemed complete by the Division prior to the effective date (July 24, 2024) on which Northern Weld County was re-classified from moderate to serious may continue to operate under this General Permit. Northern Weld County is as defined in Regulation Number 7, Part A, Section II.A.3. (Adopted 12/14/2023, Effective 02/14/2024).
- I.E. The following types of sources may not use this general permit for natural gas fired reciprocating internal combustion engines:
 - I.E.1. A portable source.
 - I.E.2. A petroleum refinery.
 - I.E.3. Newly constructed or modified existing equipment that is part of a project permit action that is subject to Non-Attainment New Source Review (NA NSR) or Prevention of Significant Deterioration (PSD) permitting under Regulation Number 3, Part D



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I.E.4. For engines newly registered or modified after Issuance 4 of this general permit, sources complying with Regulation Number 26, Part B, I.D.5.c Alternative Company-Wide Compliance Plan.

I.E.5. For engines newly registered or modified after Issuance 4 of this general permit, sources complying with Regulation Number 26, Part B, I.D.6.b.(v) alternative emission standard.

I.F. This general permit applies only to the oil and gas stationary source and the equipment as described in Section I above for which the operator has obtained approval from the Division for registration under this general permit. Other equipment at the same stationary source must be permitted separately as required by Regulation Number 3, Part B.

II. Emission Limitation Requirements

II.A. Facility-Wide Emission Limitation Requirements

II.A.1. The maximum allowable emissions of criteria air pollutants from all Air Pollutant Emission Notice (APEN)-reportable emissions points at the same stationary source (including those of the engine subject to this permit) shall be Federally Enforceable and must not exceed the following limitations. (Reference: Regulation Number 3, Part B, Section II.A.4). For facilities located outside the 8-hour Ozone Control Area (8-hour Ozone Control Area is as defined in Regulation Number 7, Part A, Section II.A.1. (Adopted 12/14/2023, Effective 02/14/2024)), facility-wide maximum allowable emissions must not exceed:

II.A.1.a.	Nitrogen Oxides (NO _x):	90.0 tons per year
II.A.1.b.	Carbon Monoxide (CO):	90.0 tons per year



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II.A.1.c. Volatile Organic Compounds (VOC): 90.0 tons per year

For sources that are also subject to Condition II.A.6. or II.A.7, the owner or operator must comply with the most stringent limits for each pollutant.

II.A.2. The maximum allowable emissions of criteria air pollutants from all APEN-reportable emissions points at the same stationary source (including those of the engine subject to this permit) shall be Federally Enforceable and must not exceed the following limitations. (Reference: Regulation Number 3, Part B, Section II.A.4). For facilities located in the 8-hour Ozone Control Area (8-hour Ozone Control Area is as defined in Regulation Number 7, Part A, Section II.A.1. (Adopted 12/14/2023, Effective 02/14/2024)), for which a complete GP02 registration request is received by the Division prior to the effective date (January 27, 2020) on which that area was re-classified from moderate to serious and are permitted under the provisions of I.B. above, facility-wide maximum allowable emissions must not exceed:

II.A.2.a. Nitrogen Oxides (NO_x): 90.0 tons per year

II.A.2.b. Carbon Monoxide (CO): 90.0 tons per year

II.A.2.c. Volatile Organic Compounds (VOC): 90.0 tons per year

For sources that are also subject to Condition II.A.6. or II.A.7., the owner or operator must comply with the most stringent limits for each pollutant.

II.A.3. The maximum allowable emissions of criteria air pollutants from all APEN-reportable emissions points at the same stationary source (including those of the engine subject to this permit) shall be Federally Enforceable and must not exceed the following limitations. (Reference: Regulation Number



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3, Part B, Section II.A.4). For facilities located in the 8-hour Ozone Control Area (8-hour Ozone Control Area is as defined in Regulation Number 7, Part A, Section II.A.1. (Adopted 12/14/2023, Effective 02/14/2024)), for which a complete GP02 registration request is not received by the Division prior to the effective date (January 27, 2020) on which that area was re-classified from moderate to serious, facility-wide maximum allowable emissions must not exceed:

II.A.3.a.	Nitrogen Oxides (NO _x):	45.0 tons per year
II.A.3.b.	Carbon Monoxide (CO):	90.0 tons per year
II.A.3.c.	Volatile Organic Compounds (VOC):	45.0 tons per year

For sources that are also subject to Condition II.A.6. or II.A.7., the owner or operator must comply with the most stringent limits for each pollutant.

- II.A.4. The maximum allowable NO_x emissions increase for the project which includes the engine covered by this general permit must be less than 40 tons per year.
- II.A.5. The maximum allowable controlled emissions of hazardous air pollutants (HAPs) from all APEN reportable emissions points at the stationary source are Federally Enforceable and must not exceed the following limitations: (Reference: Regulation Number 3, Part B, Section II.A.4.)

Each Individual HAP:	8.0 tons per year
Total of all HAPs:	20.0 tons per year

- II.A.6. The owner or operator of facilities located in an area classified as serious nonattainment for ozone, and which are not otherwise subject to



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Conditions I.B. or I.C. or I.D. of this permit, must track potential emissions from all insignificant activities at the facility on an annual basis to demonstrate compliance with facility emission limitations as indicated below. An inventory of each insignificant activity and associated emission calculations must be made available to the Division for inspection upon request. For the purposes of this condition, insignificant activities are defined as any activity or equipment, which emits any amount of VOC or NO_x but does not require an APEN or is permit exempt. Note that for reclassifications of an existing nonattainment area to serious, the limit applies upon the effective date of reclassification. (Regulation Number 3, Part B. Section III.E.)

Total point source potential emissions from the facility, including all permitted emissions and potential to emit from all insignificant activities, must be less than:

Volatile Organic Compounds (VOC): 50 tons per year

Nitrogen Oxides (NO_x): 50 tons per year

- II.A.7. The owner or operator of facilities located in an area classified as severe nonattainment for ozone, and which are not otherwise subject to Conditions I.B. or I.C. or I.D. of this permit, must track potential emissions from all insignificant activities at the facility on an annual basis to demonstrate compliance with facility emission limitations as indicated below. An inventory of each insignificant activity and associated emission calculations must be made available to the Division for inspection upon request. For the purposes of this condition, insignificant activities are defined as any activity



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or equipment, which emits any amount of VOC or NO_x but does not require an APEN or is permit exempt. Note that for reclassifications of an existing nonattainment area to severe, the limit applies upon the effective date of reclassification. (Regulation Number 3, Part B. Section III.E.)

Total point source potential emissions from the facility, including all permitted emissions and potential to emit from all insignificant activities, must be less than:

Volatile Organic Compounds (VOC): 25 tons per year

Nitrogen Oxides (NO_x): 25 tons per year

II.B. Emission Limitations Specific to Engines Registered Under this General Permit

II.B.1. All representations made in the general permit registration including, but not limited to, emission factors, nameplate and site-rated horsepower, hours of operation, emissions control measures, requested fuel consumption and other parameters that affect emissions are conditions upon which the registered engine must be constructed and operated. As such, the maximum allowable emissions from each engine registered under this general permit must not exceed the requested Federally-enforceable annual emission limitations (tons per year) as listed on the general permit registration approval letter, or if an approval letter has not been issued, as certified by the owner or operator in the APEN included with the general permit registration.

II.B.2. The owner or operator of any natural gas-fired stationary reciprocating internal combustion engine with a manufacturer's design rate greater than



500 horsepower operating in the 8-hour Ozone Control Area must comply with emission control requirements in Regulation Number 26, Part B, Section I.A., as applicable per the most current version of Regulation Number 26. (Reference: Regulation Number 26, Part B, Sections I.A.1. and I.A.2.)

- II.B.3. Engines registered to operate in accordance with the provisions of this general permit shall be subject to and shall not exceed the emissions standards contained in Table 1. If the engine is subject to a more stringent applicable performance standard for a specific pollutant, the most stringent performance standard shall apply (Reference: Regulation Number 3, Part B, Section III.E).

Table 1: Emission Standards (g/hp-hr) for Rich Burn and Lean Burn Engines

Engine Design (horsepower (hp))	NOx	CO	VOC
Rich Burn (Greater than or equal to 100 and less than 500)	2.0	4.0	1.0
Lean Burn (Greater than or equal to 100 and less than 500)	2.0	4.0	1.0
Rich Burn (Greater than or equal to 500)	0.7	1.5	1.0
Lean Burn (Greater than or equal to 500)	2.0	1.5	0.7

- II.B.4. Engines registered to operate in accordance with the provisions of this general permit with emissions from natural gas fired RICE that are greater than or equal to 100 hp and constructed in or relocated to the state of



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Colorado from another state on or after the applicable date, shall not exceed the emission standards in Table 2 as expressed in units of grams per horsepower-hour (g/hp-hr). If the engine is subject to a more stringent applicable performance standard for a specific pollutant, the most stringent performance standard shall apply (Reference: Regulation Number 3, Part B, Section III.E.).

Table 2: Emission Standards (g/hp-hr) for Rich Burn and Lean Burn Engines

Maximum Engine HP	Construction or Relocation Date	NOx	CO	VOC
Greater than or equal to 100 and less than 500	January 1, 2011	1.0	2.0	0.7
Greater than or equal to 500	July 1, 2010	1.0	2.0	0.7

- II.B.5. Engines registered to operate in accordance with the provisions of this general permit must comply with the requirements in Regulation Number 26, Part B, Section I.D.3 and Section I.D.4, as applicable per the most current version of Regulation Number 26.
- II.B.6. Engines registered to operate in accordance with the provisions of this general permit with a manufacturer's design rate greater than or equal to 1,000 horsepower must comply with the requirements in Regulation Number 26, Part B, Section I.D.5., as applicable per the most current version of Regulation Number 26.



II.B.7. Rich burn engines with a manufacturer's design rate greater than or equal to 100 horsepower but less than 1000 horsepower and lean burn engines with a manufacturer's design rate greater than or equal to 250 horsepower but less than 1000 horsepower must comply with the requirements in Regulation Number 26, Part B, Section I.D.6., as applicable per the most current version of Regulation Number 26.

II.C. Engines registered to operate in accordance with the provisions of this general permit must comply with the requirements for major sources of NO_x in Regulation Number 26, Part B, Section II.A., as applicable per the most current version of Regulation Number 26.

II.D. Air pollution control equipment may be used to comply with the emission limits and performance standards required in this general permit.

III. General Operating Conditions

III.A. The owner or operator must comply with opacity requirements in Regulation Number 1, Section II.A., as applicable per the most current version of Regulation Number 1.

III.B. This stationary source is subject to the odor requirements of Regulation Number 2.

III.C. The AIRS number (e.g. 123/1234/001) and permit number (GP02) must be marked on the subject equipment for ease of identification. (Reference: Regulation Number 3, Part B, Section III.E.)

III.D. The owner or operator must follow the Recordkeeping and Reporting/ Operating and Maintenance plans specified in Sections IV and VI. (Reference: Regulation Number 3, Part B, Section III.I.6.)



IV. Recordkeeping and Reporting

- IV.A. Records must be located on-site or at a local field office with stationary source responsibility. Records may be kept in either electronic or hard copy format provided that they can be promptly supplied to the Division upon request.
- IV.B. The following records must be maintained for the emission unit while covered by this general permit:
- IV.B.1. The current issuance of this general permit.
 - IV.B.2. The APEN submitted to the Division for each engine covered by this general permit
 - IV.B.3. The general permit registration approval letter.
 - IV.B.4. The Notice of Startup (NOS) submitted to the Division as required under Section IX.D.
- IV.C. The following records must be maintained for a period of five years:
- IV.C.1. Records that demonstrate compliance with the emission limits of this permit.
 - IV.C.1.a. Compliance Time Period
 - IV.C.1.a.(i) If the stationary source at which the engine(s) is located is a true minor source, compliance with the annual emission limits for the engine(s) must be monitored by calculating and recording the annual emissions from each engine on a calendar year basis.



IV.C.1.a.(ii) If the stationary source at which the engine(s) is located is a synthetic minor or major source, compliance with the annual emission limits for the engine(s) must be monitored by calculating emissions on a monthly basis and recording the annual emissions from each engine on a rolling twelve (12) month total. By the end of each month a new twelve-month total is calculated based on the previous twelve months' emissions data. During the first year of operation, compliance must be based upon cumulative emissions from all months of operation.

IV.C.1.b. Emissions from engines covered by this general permit must be calculated using the methodologies listed in this Section IV.C.1.b. The same emission factors, site rated horsepower, brake-specific fuel consumption (BSFC), and lower fuel heat value (LHV) as specified in the most recent registration approved by the Division must be used in the calculations. Equations Eq. 1, Eq. 2, Eq. 3 and Eq. 4 represent annual calculations and should be adjusted using the appropriate time period (monthly or annual) as required by Section IV.C.1.a.

IV.C.1.b.(i) Emission estimates based upon hours of operation must be calculated using either Eq. 1 or Eq. 2:

$$\text{Eq. 1} \quad \text{Emissions} \left(\frac{\text{tons}}{\text{yr}} \right) = EF \left(\frac{\text{g}}{\text{hp-hr}} \right) * \text{Runtime} \left(\frac{\text{hrs}}{\text{yr}} \right) * \text{Site Rated HP} * \left(\frac{\text{lbs}}{454 \text{ g}} \right) * \left(\frac{\text{ton}}{2000 \text{ lb}} \right)$$

$$\text{Eq. 2} \quad \text{Emissions} \left(\frac{\text{tons}}{\text{yr}} \right) = EF \left(\frac{\text{lb}}{\text{MMBtu}} \right) * \text{Runtime} \left(\frac{\text{hrs}}{\text{yr}} \right) * \text{Site Rated HP} * \text{BSFC} \left(\frac{\text{Btu}}{\text{hp-hr}} \right) * \left(\frac{\text{ton}}{2000 \text{ lb}} \right) * \left(\frac{\text{MMBtu}}{10^6 \text{ Btu}} \right)$$



IV.C.1.b.(ii) Emission estimates based upon fuel consumption must be calculated using either Eq. 3 or Eq. 4:

$$\text{Eq. 3} \quad \text{Emissions} \left(\frac{\text{tons}}{\text{yr}} \right) = EF \left(\frac{\text{g}}{\text{hp-hr}} \right) * \text{Fuel Use} \left(\frac{\text{scf}}{\text{yr}} \right) * \frac{1}{\text{BSFC}} \left(\frac{\text{hp-hr}}{\text{Btu}} \right) * \text{Fuel Heat Value} \left(\frac{\text{Btu}}{\text{scf}} \right) * \left(\frac{\text{lb}}{454 \text{ g}} \right) * \left(\frac{\text{ton}}{2000 \text{ lb}} \right)$$

$$\text{Eq. 4} \quad \text{Emissions} \left(\frac{\text{tons}}{\text{yr}} \right) = EF \left(\frac{\text{lb}}{\text{MMBtu}} \right) * \text{Fuel Use} \left(\frac{\text{scf}}{\text{yr}} \right) * \text{Fuel Heat Value} \left(\frac{\text{Btu}}{\text{scf}} \right) * \left(\frac{\text{MMBtu}}{10^6 \text{ Btu}} \right) * \left(\frac{\text{ton}}{2000 \text{ lbs}} \right)$$

IV.C.2. Records of the hours of operation and fuel consumption for each engine covered by this permit must be recorded and made available to the Division upon request. Records must be for the same time period (monthly or annual) as determined by Section IV.C.1.a. for the emission calculations. Fuel consumption may be individually metered, allocated from a master meter, or estimated using Eq. 5 based upon the site-rated horsepower, brake-specific fuel consumption (BSFC), and the fuels lower fuel heat value (LHV) as specified in the most recent registration approved by the Division.

$$\text{Eq. 5} \quad \text{Fuel Consumption} \left(\frac{\text{scf}}{\text{yr}} \right) = \text{BSFC} \left(\frac{\text{Btu}}{\text{hp-hr}} \right) * \text{Site Rated HP} * \text{Runtime} \left(\frac{\text{hr}}{\text{yr}} \right) * \frac{1}{\text{Fuel Heat Value}} \left(\frac{\text{scf}}{\text{Btu}} \right)$$

IV.C.3. An alternative operating scenario (AOS) modification log containing all details required by any AOS in this permit. (Reference: Regulation Number 3, Part A, Section IV.A.1.)

IV.C.4. Records that demonstrate compliance with the facility-wide emission limit requirements that facilities must meet in order to qualify for use of this General Permit as set forth in Section II.A. of this permit.



IV.C.5. Results of initial compliance and periodic performance testing required by Sections V and VI.

IV.C.6. Operating and maintenance records required by Section VI.

IV.D. On or before June 30th of each calendar year, the owner or operator of oil and natural gas operations and equipment at or upstream of a natural gas processing plant or class II disposal well facilities in Colorado 02 must submit a single annual report that includes actual emissions and specified information in the Division-approved report format. The information included in the annual report must be in accordance with the general reporting requirements of Regulation Number 7, Part B, Section V.B. Each calendar year, owners or operators must maintain the information according to Regulation Number 7, Part B, Section V.C. for inclusion in the annual report. (Reference: Regulation Number 7, Part B, Section V).

IV.E. Engines registered under GP02 and subject to the requirements of Regulation Number 26, Part B, Section I.D.5 must comply with the recordkeeping and reporting requirements listed in Regulation Number 26, Part B, Sections I.D.5.f. and I.D.5.g.

IV.F. Engines registered under GP02 and subject to the requirements of Regulation Number 26, Part B, Section I.D.6 must comply with the recordkeeping and reporting requirements listed in Regulation Number 26, Part B, Sections I.D.6.e. and I.D.6.f.

V. Compliance Test Requirements

V.A. If the stationary source is synthetic minor or major for NO_x or CO an initial compliance test for outlet concentrations of NO_x and CO is required. If the stationary source is synthetic minor for formaldehyde, an initial compliance test for outlet concentrations of NO_x, CO and formaldehyde is required. Initial



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compliance test requirements only apply to engines equipped with a catalytic emission control device that are nameplate-rated greater than or equal to 500 hp.

- V.B. Engines registered under GP02 and subject to the requirements of Regulation Number 26, Part B, Section I.D.5. must comply with the performance testing requirements listed in Regulation Number 26, Part B, Section I.D.5.d.
- V.C. Engines registered under GP02 and subject to the requirements of Regulation Number 26, Part B, Section I.D.6. must comply with the performance testing requirements listed in Regulation Number 26, Part B, Section I.D.6.c.
- V.D. The Division reserves the right to require a compliance test on any engine registered under this general permit. (Reference: Regulation Number 3, Part B, Section III.E)
- V.E. Results from all compliance tests must be reported in units of parts per million dry volume (ppmv), grams per horsepower hour (g/hp-hr) and pounds per million Btu (lb/MMBtu). In order to demonstrate compliance with the registered engine's individual annual emissions limits established in Section II.B.1, results of the test must be converted to a pound per hour (lb/hr) basis and multiplied by the allowable operating hours in a year.
- V.F. All initial compliance tests required by this section must meet the following conditions:
 - V.F.1. For new engines, the initial compliance test must be completed within 180 days of commencement of operation of the subject engine.
 - V.F.2. For existing engines, the initial compliance test must be completed within 180 days of registering the subject engine(s), unless required earlier by condition V.B or V.C. A compliance test performed within the previous five



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years prior to registration that was used to fulfill a previous Colorado permitting requirement may be used to comply with the initial compliance test requirement providing all other conditions are met.

- V.F.3. The test protocol must be in prepared in accordance with the requirements of the Division Compliance Test Manual and must be submitted to the Division for review and approval at least thirty (30) days prior to testing. If after thirty (30) calendar days, the owner or operator has not received approval and the Division did not request additional information from the owner or operator, the protocol will be considered automatically approved after the 30-day period.
- V.F.4. The Division must be notified at least 30 days prior to the test date. The owner or operator may request a waiver of this 30-day notification requirement subject to approval of the Division.
- V.F.5. Results of all initial compliance tests completed to fulfill the requirements of this Section V must be submitted to the Division for review within thirty (30) days of completion of the test.

VI. Engine Operating and Maintenance Plan

- VI.A. This Section VI applies only to engines covered by this general permit that are located at a stationary source that is synthetic minor or major for NO_x, CO, or formaldehyde except as indicated in Conditions VI.E.1, VI.E.2, and VI.E.6.
- VI.B. These general requirements apply to all subparts of Section VI.
 - VI.B.1. Each engine and associated catalytic control device, if applicable, must be maintained and operated according to the manufacturer's scheduled maintenance guide or equivalent. A company may establish its own



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equivalent maintenance schedule providing it achieves efficient engine operation and meets or exceeds the performance standards of this permit.

VI.B.2. A copy of maintenance schedules must be kept.

VI.B.3. Records indicating the date and description of maintenance must be kept.

VI.B.4. If maintenance activities or actions are dependent upon hours of operation, then engine operating hours must be tracked and recorded. If a method other than hours of operation is used, then the tracking method must be recorded.

VI.C. For each rich burn engine controlled with a NSCR converter the following conditions must be met:

VI.C.1. The engine must be equipped with an air-fuel ratio controller (AFRC).

VI.C.2. Each AFRC must be maintained per the manufacturer's recommended maintenance or equivalent, including replacement of the oxygen sensor as necessary for oxygen sensor-based controllers. In the absence of specific manufacturer's recommendations or equivalent, the oxygen sensor must be replaced quarterly (or within a 2190 hour operating period if the engine does not run continuously).

VI.D. For each lean burn engine equipped with a NO_x sensor the operator must follow the manufacturer's recommended maintenance schedule or equivalent.

VI.E. For each engine equipped with a catalytic emission control device, the following conditions must be met:



- VI.E.1. Engines subject to the requirements of Regulation Number 26, Part B, Section I.D.5. must comply with the monitoring requirements listed in Regulation Number 26, Part B, Section I.D.5.e.)
- VI.E.2. Engines subject to the requirements of Regulation Number 26, Part B, Section I.D.6. must comply with the monitoring requirements listed in Regulation Number 26, Part B, Section I.D.6.d.
- VI.E.3. The catalyst must be cleaned, reconditioned and replaced per the manufacturer's recommended maintenance or equivalent.
- VI.E.4. The catalyst inlet temperature must be recorded weekly, unless a more stringent requirement is required in Regulation Number 26, Part B, Section I.D.5.e.(ii). If the temperature is outside of the range specified in Section VI.E.4.a or VI.E.4.b, corrective action must be taken and documented.
 - VI.E.4.a. For a rich burn engine with a non-selective catalytic reduction (NSCR) emission control device, the temperature into the catalyst must be between 750°F and 1250°F.
 - VI.E.4.b. For a lean burn engine with an oxidation catalyst emission control device, the temperature into the catalyst must be between 450°F and 1350°F.
- VI.E.5. The pressure drop across the catalyst bed must be recorded monthly. The monthly pressure drop values must not deviate by more than +/- 2 inches of water column of the most recent baseline value established according to Section VI.E.5.b.



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VI.E.5.a. If the pressure drop value is out of range, the owner or operator must perform any one or more of the following:

VI.E.5.a.(i) Corrective action must be taken and documented, or

VI.E.5.a.(ii) The owner or operator may artificially load the engine to within +/- 10 percent of the most recent baseline load value established. If the pressure drop is then within range, the pressure drop and load values must be documented and no further corrective action is required, or

VI.E.5.a.(iii) The owner or operator may forego maintenance activities and perform a portable analyzer test to establish a new pressure drop baseline.

VI.E.5.b. The baseline pressure drop must be established by recording the engine load and pressure drop across the catalyst during engine testing. The baseline pressure drop must be periodically reestablished as follows.

VI.E.5.b.(i) During each initial compliance test required by Section V. If historical initial compliance test data is used to fulfill Section V and baseline pressure drop was not recorded during the compliance test, the operator may defer



establishing the baseline pressure drop until the first periodic performance test required by Section VI.E.5.

VI.E.5.b.(ii) During each periodic performance test required by Section VI.E.5.

VI.E.5.b.(iii) Within the first thirty (30) days of engine operation from the date that the catalyst is cleaned, reconditioned, or replaced.

VI.E.6. For each engine equipped with a catalytic emission control device, periodic performance tests for NO_x and CO using a portable analyzer are required to indicate continued performance of the catalyst and proper operation of the engine as follows:

VI.E.6.a. Semi-annually at synthetic minor facilities (or within a 4,380 hour operating period if the engine does not run continuously); and

VI.E.6.b. Effective June 1, 2020 semi-annually at true minor facilities (or within a 4,380 hour operating period if the engine does not run continuously); and

VI.E.6.c. Effective June 1, 2020 quarterly at major facilities (or within a 2,190 hour operating period if the engine does not run continuously)

VI.E.6.d. For engines that do not operate continuously and which do not exceed 4,380 hours of operation within two (2)



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calendar years, a minimum of one performance test must be completed every two years.

- VI.E.6.d.(i) Each performance test should be performed with the engine operating at the highest load practicable. However, it is not required to artificially load the engine in order to perform the test.
- VI.E.6.d.(ii) All performance tests using a portable analyzer, including tests triggered by a replacement engine authorized by an AOS in this permit, must meet the requirements established in the Portable Analyzer Monitoring Protocol or other Division approved protocol.
- VI.E.6.d.(iii) At the owner or operator's discretion, a performance test using a reference test method may be used instead of a portable analyzer.

VII. Engine Alternative Operating Scenarios (AOS)

VII.A. Provided that the emission limits set forth in Section II are still met, the owner or operator may invoke an AOS for the following modifications to an existing engine registered under the general permit:

- VII.A.1. An existing registered engine may be temporarily replaced with a different or like-kind engine in accordance with Attachment A for "Alternative Operating Scenarios Reciprocating Internal Combustion Engines".



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VII.A.1.a. A temporary replacement is defined as in the same service for:

VII.A.1.a.(i) 270 operating days or less in any 12-month period for sources subject to Section VII.D. or

VII.A.1.a.(ii) 90 operating days or less in any 12-month period for sources not subject to Section VII.D..

The number of days is the total number of days that the equipment is in operation. If the equipment operates only part of a day, that day counts towards a full day towards the total.

VII.A.1.b. A like-kind replacement is defined as replacing a piece of equipment with the same make, model, and design capacity as the original piece of equipment.

VII.A.1.c. Temporary replacements allowed under the AOS in this permit do not require a revised APEN or registration to be submitted.

VII.A.2. An existing registered engine may be permanently replaced with a like-kind engine in accordance with Attachment A for “Alternative Operating Scenarios Reciprocating Internal Combustion Engines”.

VII.A.2.a. A permanent replacement is defined as in the same service for more than 90 operating days in any 12 month period. The 90 days is the total number of days that the equipment is in operation. If the equipment



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operates only part of a day, that day counts towards the 90 day total.

- VII.A.2.b. A like-kind replacement is defined as replacing a piece of equipment with the same make, model, and design capacity as the original piece of equipment.
- VII.A.2.c. The compliance demonstrations and any periodic monitoring required by this AOS are in addition to any compliance demonstrations or periodic monitoring required by this permit.
- VII.A.2.d. An APEN that includes the specific manufacturer, model and serial number and horsepower of the permanent replacement engine must be filed with the Division for the permanent replacement engine within 14 calendar days of commencing operation of the replacement engine. The APEN shall be accompanied by the appropriate APEN filing fee, GP registration fee, a cover letter explaining that the permittee is exercising an alternative operating scenario and is installing a permanent replacement engine, and a copy of the relevant Applicability Reports for the replacement engine.
- VII.A.2.e. Permanent replacement engines must comply with portable analyzer testing requirements set forth in Attachment A, Section 6.



VII.A.2.f. A permanent replacement engine must not include changes to the configuration that might impact a dispersion analysis, this includes but is not limited to:

- Stack heights;
- Stack locations; or
- Exit velocity

VII.A.3. Routine maintenance, repair, or like-kind replacement of control equipment

VII.A.4. Addition of a control device where the operator will not claim Federally enforceable credit for the emission reductions achieved.

VII.B. Modified or replacement equipment are subject to all requirements set forth in this permit including, but not limited to, recordkeeping and reporting, and operation and maintenance plans.

VII.C. An AOS otherwise allowed by this general permit may not be used if the modification will result in a facility classification change from true minor to synthetic minor. In this case, a revised registration request must be submitted for review.

VII.D. An AOS otherwise allowed by this general permit cannot be used for the permanent replacement of an entire engine at any source that is currently a major stationary source for purposes of Prevention of Significant Deterioration or Non-Attainment Area New Source Review (“PSD/NANSR”) unless the existing engine has emission limits that are below the significance levels in Regulation Number 3, Part D, II.A.42. Nothing in this AOS shall preclude the Division from taking an action, based on any permanent engine replacement(s), for circumvention of any state or



federal PSD/NANSR requirement. Additionally, in the event that any permanent engine replacement(s) constitute(s) a circumvention of applicable PSD/NANSR requirements, nothing in this AOS shall excuse the permittee from complying with PSD/NANSR and applicable permitting requirements.

- VII.E. The owner or operator must maintain a log to contemporaneously record a description of all modifications made under the provisions of this AOS. Information recorded for temporary engine replacements must include: the start and stop date of any engine replacement, the manufacturer, model number, serial number, and design rating of both the engine replaced and the replacement engine. Records for emission control equipment additions must include the date of installation of new emissions control equipment. (Reference: Regulation Number 3, Part A, Section IV.A.1.)
- VIII. Monitoring Requirements for Facilities located in Disproportionately Impacted (DI) Communities**
- VIII.A. Owners or operators of new or modified sources located in Disproportionately Impacted Communities must submit a notification at the time of registration for a general permit or permits issued pursuant to Regulation Number 3, Part B, Section III.I. indicating the source will comply with the monitoring requirements outlined in Regulation Number 3, Part B, Section III.J., as applicable, by either:
- VIII.A.1. Complying with source-specific monitoring pursuant to Section III.J.2. of Part B;
 - VIII.A.2. Paying community monitoring fees pursuant to Section III.J.3. of Part B; or



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VIII.A.3. Complying with well production facility monitoring pursuant to Section III.J.4. of Part B.

Owners or operators must comply with the monitoring methodology selected on the general permit registration form.

VIII.B. Owners or operators subject to VIII.A.1 must submit a Division-approved source-specific monitoring plan as part of a complete registration request. (Regulation Number 3, Part B, III.E.)

IX. Additional Requirements

IX.A. Engines subject to requirements of 40 CFR Part 60, Subpart JJJJ, Standards of Performance for Stationary Spark Ignition Internal Combustion Engines, must comply with all applicable requirements of that subpart including, but not limited to, emissions standards, testing, notification, monitoring, records and reporting.

IX.B. Engines subject to area source requirements of 40 CFR Part 63, Subpart ZZZZ, National Emissions Standards for Hazardous Air Pollutants for Reciprocating Internal Combustion Engines, must comply with all applicable requirements of that subpart including, but not limited to, performance standards, testing, notification, monitoring, records and reporting.

IX.C. The owner or operator must maintain documentation clearly outlining applicable requirements of New Source Performance Standard (NSPS) Subpart JJJJ and Maximum Achievable Control Technology (MACT) Subpart ZZZZ for the engine registered under this permit. The documentation must be located on-site or at a local field office with stationary source responsibility. Records may be kept in either electronic or hard copy format provided that they can be promptly supplied to the Division upon request.



- X. General Permit Registration Approval Process**
- X.A. Conditional certification of a registration under this general permit is effective from the date the registration is deemed complete as indicated in Condition X.B. A complete registration request consists of all General Permit application materials required by the Division and associated fees. The owner or operator may commence construction and operation of the engine as represented in the registration once the registration is deemed complete. In the event the engine does not qualify for registration under the general permit the owner or operator accepts the liability of commencing these activities.
- X.B. The Division will determine completeness of the General Permit registration request within sixty (60) days of the date of receipt of the request and provide written notification to the applicant. If the applicant does not receive notification of a completeness determination from the Division within sixty (60) days, the General Permit registration request will be deemed complete for the purpose of Conditions X.A and X.B.
- X.C. The Division has ninety (90) calendar days from the date the Division receives a complete General Permit registration request to provide the applicant with a written approval or denial of the registration.
- X.D. The owner or operator must submit a Notice of Startup (NOS) within fifteen (15) calendar days after the commencement of operation of the engine registered to this general permit. The notice of startup must include the date of startup and make, model, and serial number of the registered engine. If all required information is available at the time of registration, the notice of startup may be submitted with the General Permit registration request. The NOS must be submitted using the Division's online submission tool on the Division's self-



certification webpage at <https://cdphe.colorado.gov/apens-and-air-permits/air-permit-self-certification>.

XI. General Permit Terms and Administration

XI.A. A facility that is classified as a major source for Title V Operating Permit requirements and is operating under Condition I.A.4 or I.A.5 must submit a Title V operating permit application requesting to incorporate the terms of this general permit into a Title V Operating Permit. The due date for this application is one year from the date on which the facility becomes subject to Title V Operating Permit requirements.

XI.B. In accordance with Colorado Regulation Number 3, Part C, Section V.C.5, some or all of the monitoring specified in this general permit will be incorporated as specific conditions in the source's Title V Operating Permit (if applicable). Additional monitoring not listed in this general permit may be included in the source's Operating Permit in order to satisfy the periodic monitoring requirements of Section V.C.5.b.

XI.C. General Terms

XI.C.1. A revised APEN shall be filed: (Reference: Regulation Number 3, Part A, Section II.C.)

XI.C.1.a. Annually by 30 April of the year following a significant increase in emissions as follows:

XI.C.1.a.(i) A change in criteria pollutant actual emissions of five (5) tons per year or more, above the level reported on the last APEN submitted to the Division; or



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- XI.C.1.a.(ii) For volatile organic compounds and nitrogen oxides sources in ozone non-attainment areas, a change in annual actual emissions of one ton per year or more or five percent, whichever is greater, above the level reported on the last APEN submitted to the Division; or
- XI.C.1.a.(iii) For any non-criteria reportable pollutant, if the emissions increase by 50% or five (5) tons per year, whichever is less, above the level reported on the last APEN.
- XI.C.1.b. Whenever there is a change in the owner or operator of any facility, process, or activity; or
- XI.C.1.c. As required by the alternative operating scenario allowed by Section III.
- XI.C.1.d. No later than 30 days before the existing APEN expires.
- XI.C.2. Certain requirements of Regulation Number 3, Part D shall apply at such time that any stationary source or modification becomes a major stationary source or major modification solely by virtue of a relaxation in any enforceable limitation that was established after August 7, 1980 on the capacity of the source or modification to otherwise emit a pollutant, such as a restriction on hours of operation. (Reference: Regulation Number 3, Part D, § V.A.7.b. and VI.B.4.)
- XI.C.3. This stationary source may be able to utilize the Affirmative Defense Provision for Excess Emissions During Malfunctions contained in Common Provisions Regulation Part II, Subpart E. The owner or operator must notify



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the Division of any malfunction condition which causes a violation of any emission limit or limits stated in this permit as soon as possible, but no later than noon of the next working day, followed by written notice to the Division addressing all of the criteria set forth in Part II.E.1 of the Common Provisions Regulation.

- XI.C.4. This permit is granted subject to all rules and regulations of the Colorado Air Quality Control Commission and the Colorado Air Pollution Prevention And Control Act C.R.S., sections 25-7-101, et seq, to those general and specific terms and conditions included in this document.
- XI.C.5. Unless specifically stated otherwise, the general and specific conditions contained in this permit have been determined by the Division to be necessary to assure compliance with the provisions of Section 25-7-114.5(7)(a), C.R.S.
- XI.C.6. Each and every condition of this permit is a material part hereof and is not severable. Any challenge to or appeal of a condition herein by the permittee will constitute a rejection of the entire permit and upon such occurrence, this permit must be deemed denied ab initio.
- XI.C.7. Section 25-7-114.7(2)(a), C.R.S. requires that all sources required to file an APEN must pay an annual fee to cover the costs of inspections and administration.
- XI.C.8. Violation of the terms of a permit or of the provisions of the Colorado Air Pollution Prevention and Control Act or the regulations of the AQCC may result in administrative, civil or criminal enforcement actions under Sections 25-7-115 (enforcement), -121 (injunctions), -122 (civil penalties), -122.1 (criminal penalties), C.R.S.



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- XI.C.9. Registration under this permit is approved in reliance upon the accuracy and completeness of information supplied by the applicant and is conditioned upon operation of the source, in accordance with this information and with representations made by the applicant or applicant's agents. It is valid only for the equipment and operations or activity specifically identified on the general permit registration.
- XI.C.10. All terms and conditions of this permit that apply to the equipment covered by this permit must be considered Applicable Requirements for the purposes of any future permit issued for the engine(s). (Reference: Regulation Number 3, Part A, Section I.B.9.a)

XI.D. Registration Revision / Termination

- XI.D.1. The Division may deny or revoke registration under the general permit under the circumstances specified in Regulation Number 3, Part B, Section III.I.3.c.
- XI.D.2. A registration under this general permit may be reissued to a new owner or in a new company name by the Division as provided in Regulation Number 3, Part B, Section II.B. upon a request for transfer of ownership or company name change and the submittal of a revised APEN and the required fees.
- XI.D.3. Registration under this general permit is voluntary. The owner or operator may withdraw or cancel a registration under this general permit at any time by notifying the Division in writing and applying for an individual permit as required by Regulation Number 3, Part B.

XI.E. General



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- XI.E.1. This general permit remains in effect until revised or terminated by the Division in accordance with the provisions of Regulation Number 3.
- XI.E.2. After public notice and comment as provided by Regulation Number 3, Part B, Section III.I.7., the Division may revise this general permit in order to add or delete requirements or limitations to the permit. This public notice shall be conducted in a manner consistent with the provisions of Regulation Number 3, Part B, Section III.C.4.
- XI.E.3. If the Division revises this general permit, it will provide written notice to affected owner or operator prior to the revision of the general permit. The notice will advise the owner or operator that any existing registration to use the general permit will be automatically converted to a registration to use the revised general permit, and the owner or operator is subject to requirements of the revised general permit. Persons not wishing to continue coverage under the revised general permit shall have the option of applying for an individual permit as required by Regulation Number 3, Part B.
- XI.E.4. If the Division terminates this general permit, it will provide written notice to affected registrants prior to the termination of the general permit. The notice will advise registrants that they must apply for an individual permit as required by Regulation Number 3, Part B.
- XI.E.5. The Division may require any source authorized by a general construction permit to apply for and obtain an individual permit if circumstances have changed since the time of the original general permit application so that the source is no longer appropriately controlled and/or permitted under the general construction permit. (Reference: Regulation Number 3, Part B, § III.I.3.c.(i)(B).



Permit History

Issuance	Date	Description
Issuance 1	1 August 2011	Initial issuance
Issuance 2	January 24, 2020	Revised to address ozone non-attainment area re-classification from moderate to serious.
Issuance 3	November 3, 2022	Revised to address the 8-hour Ozone Control Area re-classification from serious to severe for existing registrations
Issuance 4	This Issuance	Revised to address: • Revised to address Northern Weld County Area re-classification from moderate to serious for existing registrations. • Additional changes due to reclassification from serious to severe non-attainment • Additional changes due to reclassification from moderate to serious non-attainment. • Changes to the structure of Air Quality Control commission (AQCC) Regulations Number 7 and Number 26; • Transfer of requirements from AQCC Regulation Number 7 to AQCC Regulation Number 26.



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Issuance	Date	Description
		• New requirements applicable to engines with rating greater than or equal to 1000 hp. • Updates to Operating and Maintenance requirements. • Addition of a permanent AOS



Attachment A: 2/14/2024

Alternative Operating Scenarios

The following Alternative Operating Scenario (AOS) for the temporary and permanent replacement of natural gas fired reciprocating internal combustion engines has been reviewed in accordance with the requirements of Regulation Number 3., Part A, Section IV.A, Operational Flexibility-Alternative Operating Scenarios, Regulation Number 3, Part B, Construction Permits, and Regulation Number 3, Part D, Major Stationary Source New Source Review and Prevention of Significant Deterioration, and it has been found to meet all applicable substantive and procedural requirements. This permit incorporates and is considered to be a Construction Permit for any engine replacement performed in accordance with this AOS, and the owner or operator is allowed to perform such engine replacement without applying for a revision to this permit or obtaining a new Construction Permit.

The following AOS is incorporated into this permit in order to deal with an engine breakdown or periodic routine maintenance and repair of an existing onsite engine that requires the use of either a temporary or permanent replacement engine. The compliance demonstrations and any periodic monitoring required by this AOS are in addition to any compliance demonstrations or periodic monitoring required by this permit.

1. Prohibitions

- 1.1. The AOS **cannot be used** for the permanent replacement of an entire engine at any source that is currently a major stationary source for purposes of Prevention of Significant Deterioration or Non-Attainment Area New Source Review ("PSD/NANSR") unless the existing engine has emission limits that are below the significance levels in Regulation Number 3, Part D, II.A.44. Nothing in this AOS precludes the Division from taking an action, based on any permanent engine replacement(s), for circumvention of any state or federal PSD/NANSR requirement. Additionally, in the event that any permanent engine replacement(s) constitute(s) a circumvention of applicable PSD/NANSR requirements, nothing in this AOS excuses the owner or operator from complying with PSD/NANSR and applicable permitting requirements.



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- 1.2. This AOS **cannot be used** to change the annual permitted emission limits or any compliance emission factor or calculation method that is required by a permit to be used for the purposes of monitoring compliance with the annual permitted emission limits.
- 1.3. This AOS **cannot be used** for the permanent replacement of an engine subject to an emission limitation under Regulation Number 26, Part B, Section I.D.5 (state-wide requirements for natural gas-fired reciprocating internal combustion engines with manufacturer's design rate greater than or equal to 1000 hp), unless the replacement is subject to the same numerical emission standard and compliance date as the unit being replaced, on the date that the replacement will occur.
- 1.4. This AOS **cannot be used** for the permanent replacement of an engine subject to an emission limitation under Regulation Number 26, Part B, Section I.D.6 (state-wide requirements for rich burn and lean burn natural gas-fired reciprocating internal combustion engines with manufacturer's design rate greater than or equal to 100 hp and 250 hp, respectively, and less than 1000 hp), unless the replacement is subject to the same numerical emission standard and compliance date as the unit being replaced, on the date that the replacement will occur.
- 1.5. Additional Sources

The replacement of an existing engine with a new engine is viewed by the Division as the installation of a new emissions unit, not “routine replacement” of an existing unit. The AOS is therefore essentially an advanced construction permit review. The AOS cannot be used for additional new emission points for any site; an engine that is being installed as an entirely new emission point and not as part of an AOS-approved replacement of an existing onsite engine has to go through the appropriate Construction/Operating permitting process prior to installation.

2. General Provisions

All replacement engines are subject to all federally applicable and state-only requirements set forth in this permit for the replaced engine (including monitoring and recordkeeping), *unless*



otherwise superseded by the applicable requirements identified in the Applicability Report for a particular permanently replaced engine.

The results of all tests and the associated calculations required by this AOS must be submitted to the Division within 30 calendar days of the test or within 60 days of the test if such testing is required to demonstrate compliance with NSPS or MACT requirements. Results of all tests must be maintained and made available to the Division upon request.

2.1. Contemporaneous Log

The owner or operator must maintain a log on-site and contemporaneously record the start and stop date of any engine replacement, the manufacturer, date of manufacture, model number, horsepower, and serial number of the engine(s) that are replaced during the term of this permit, and the manufacturer, model number, horsepower, and serial number of the replacement engine. In addition to the log, for permanent engine replacements, the owner or operator must maintain a copy of all Applicability Reports required under Item 4 below and make them available to the Division upon request.

2.2. Definition of Temporary Replacement and Permanent Replacement

“Temporary” is defined as in the same service for either 90 operating days or 270 operating or less in any 12 month period, based on the total number of days that the engine is in operation. If the engine operates only part of a day, that day must count as a single day towards the total. Except as provided below for natural gas-fired reciprocating internal combustion engines subject to an emission limitation under Regulation Number 26, Part B, Section I.D.5 or I.D.6., temporary is defined as 90 operating days for engines where permanent replacements are allowed or 270 operating days where permanent replacements are not allowed.

“Temporary” for natural gas-fired reciprocating internal combustion engines subject to an emission limitation under Regulation Number 26, Part B, Section I.D.5 or I.D.6. is defined as 270 days operating days.



“Permanent” is defined as in the same service for more operating days than allowed under temporary replacement, as defined for each specific engine.

3. Specific Provisions for Temporary Replacements

The owner or operator may temporarily replace an existing engine that is subject to the emission limits set forth in this permit with an engine that is of the same manufacturer, model, and horsepower or a different manufacturer, model, or horsepower as the existing engine without modifying the general permit registration, so long as the temporary replacement engine complies with all permit limitations and other requirements applicable to the existing engine. The owner or operator must also comply with any additional requirements applicable to the temporary replacement. Measurement of emissions from the temporary replacement engine must be made as set forth in Item 6 below.

4. Specific Provisions for Permanent Replacements

Except as provided in Item 1 above, the owner or operator may permanently replace an existing engine with another engine with the same manufacturer, model, and horsepower without modifying the general permit registration so long as the permanent replacement engine complies with all permit limitations and requirements applicable to the existing engine, unless otherwise superseded by the regulations identified in the Applicability Report, as well as any other applicable requirements for the replacement engine. Measurement of emissions from the permanent replacement engine and compliance with the applicable emission limitations must be made as set forth in Item 6 below.

An Air Pollutant Emissions Notice (APEN) that includes the specific manufacturer, model and serial number and horsepower of the permanent replacement engine must be filed with the Division for the permanent replacement engine within 14 calendar days of commencing operation of the replacement engine. The annual emission limitations and associated emission factors requested on the APEN must exactly match the annual emission limitations and emission factors associated with the annual emission limits set forth in the permit for the existing engine. Any changes to these values cannot be processed as a permanent replacement under this AOS and will instead require a permit modification.



The APEN must be accompanied by the appropriate APEN filing fee, a cover letter explaining that the owner or operator is exercising an alternative operating scenario and is installing a permanent replacement engine, and a copy of the relevant Applicability Reports for the replacement engine. Applicability Reports can be found at <https://cdphe.colorado.gov/alternate-operating-scenario-aos-for-engines>. This submittal must be accompanied by a certification from the person legally authorized to act on behalf of the owner or operator indicating that “based on the information and belief formed after reasonable inquiry, the statements and information included in the submittal are true, accurate and complete”.

Annual emission calculations for monitoring compliance with annual permit limits for permanent replacement engines must be performed using the emission factors and calculation methods set forth in the permit for the existing engine. If the owner or operator elects to modify either the annual emission limit or the emission factor in the permit associated with annual limit to more closely reflect the replacement engine operation, a permit modification application must be submitted to the Division. The use of any emission factors or calculation methods for the purposes of monitoring compliance with annual limits is prohibited until a permit application has been processed by the Division and a permit revision issued.

Nothing in this AOS is intended to require the owner or operator to use emission factors associated with annual permit limits to monitor compliance with any other emission limits or standards, unless the owner or operator is already explicitly required to do so by a specific permit condition or applicable requirement.

The owner or operator must pay fees based on the normal permit processing rate for review of information submitted to the Division in regard to any permanent engine replacement.

5. Applicable Regulations for Permanent Engine Replacements

An analysis of all applicable requirements, including but not limited to monitoring, recordkeeping and reporting, for the permanent engine replacement must be included in the Applicability Reports required under Item 4. Any testing required by the rules listed in items 5.2 through 5.9 below is in addition to that required by this AOS.



5.1. Reasonably Available Control Technology (RACT): Reg 3, Part B § III.D.2

All permanent replacement engines that are located in an area that is classified as attainment/maintenance or nonattainment must apply Reasonably Available Control Technology (RACT) for the pollutants for which the area is attainment/maintenance or nonattainment. Note that both VOC and NO_x are precursors for ozone. RACT must be applied for any level of emissions of the pollutant for which the area is in attainment/maintenance or nonattainment, except as follows:

In the Denver Metropolitan PM₁₀ attainment/maintenance area, RACT applies to PM₁₀ at any level of emissions and to NO_x and SO₂, as precursors to PM₁₀, if the potential to emit of NO_x or SO₂ exceeds 40 tons/yr.

For purposes of this AOS, the following are considered RACT for natural-gas fired reciprocating internal combustion engines:

VOC: The emission limitations in NSPS JJJJ

CO: The emission limitations in NSPS JJJJ

NO_x: The emission limitations in NSPS JJJJ

SO₂: Use of natural gas as fuel

PM₁₀: Use of natural gas as fuel

As defined in 40 CFR Part 60 Subparts GG (§ 60.331) and 40 CFR Part 72 (§ 72.2), natural gas contains 20.0 grains or less of total sulfur per 100 standard cubic feet.

5.2. Regulation Number 26, Part B, Section I.A - Requirements for new and existing engines (in the 8-hour Ozone Control Area)

Natural gas-fired stationary or portable reciprocating internal combustion engines in the 8-hour Ozone Control Area with manufacturer's design rate greater than 500 horsepower shall



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employ a non-selective catalyst reduction system and air fuel controller (for rich burn engines), or an oxidation catalyst (for lean burn engines).

For specific rule requirements, see:

Section I.A Applicability and compliance dates

Section I.B Requirements for Control Devices

Section I.C Exemptions

5.3. Regulation Number 26, Part B, Section I.D.3 - New, Modified and Relocated Natural Gas Fired Reciprocating Internal Combustion Engines (State Only)

A natural gas-fired reciprocating internal combustion engines that is either constructed or relocated to the state of Colorado from another state after the date listed in the table are is subject to the emission standards in the table below (labeled as Table 1 in the Part B).

Max Engine (HP)	Construction or Relocation Date	NO _x (g/hp-hr)	CO (g/hp-hr)	VOC (g/hp-hr)
100≤hp<500	January 1, 2008	2.0	4.0	1.0
	January 1, 2011	1.0	2.0	0.7
500≤hp	July 1, 2007	2.0	4.0	1.0
	July 1, 2010	1.0	2.0	0.7

For specific rule requirements, see:

Section I.D.1 Exemptions

Section I.D.2 General Provisions



Section I.D.3 Emission Standards

5.4. Regulation Number 26, Part B, Section I.D.4 - Existing Natural Gas Fired Reciprocating Internal Combustion Engines

Rich Burn Natural gas-fired reciprocating internal combustion engines with manufacturer's name plate design rate greater than 500 horsepower constructed or modified before February 1, 2009 shall install and operate a non-selective catalyst reduction system and air fuel controller.

(State-Only) All lean Burn Natural gas-fired reciprocating internal combustion engines with manufacturer's name plate design rate greater than 500 horsepower shall install and operate an oxidation catalyst.

For specific rule requirements, see:

Section I.D.1	Exemptions
Section I.D.4.a.(i)(B)-(C)	Exemptions
Section I.D.4.a.(i)(A)	General Provisions
Section I.D.4.a.(i)	Control Requirements
Section I.D.4.b.(i)	Control Requirements
Section I.D.4.c.	Control Requirements listed engines $\geq$ 1000 HP

5.5. Additional Requirements for Natural Gas Fired Reciprocating Internal Combustion Engines: Regulation Number 26, Part B, Section I.D.5 (State-Only).

Stationary natural gas-fired reciprocating internal combustion engines with a manufacturer's design rate greater than or equal to 1000 horsepower are subject to specific emission standards. The specific applicable standards and compliance date of any particular engine or any particular replacement of that engine at any particular point in time is variable,



dependent on multiple factors, and can potentially be subject to further changes at the discretion of the owner or operator.

For specific rule requirements, see:

- Section I.D.5.a Applicability and exemptions
- Section I.D.5.b Standards for engines not under a company-wide plan
- Section I.D.5.c Alternative Company-Wide Compliance Plan
- Section I.D.5.d Performance Testing
- Section I.D.5.e Monitoring and Combustion Process Adjustment
- Section I.D.5.f Recordkeeping
- Section I.D.5.g Reporting

5.6. Additional Requirements for Natural Gas Fired Reciprocating Internal Combustion Engines:
Regulation Number 26, Part B, Section I.D.6 (State-Only).

Rich burn stationary natural gas-fired reciprocating internal combustion engines with a manufacturer's design rate greater than or equal to 100 horsepower but less than 1000 horsepower and lean burn stationary natural gas-fired reciprocating internal combustion engines with a manufacturer's design rate greater than or equal to 250 horsepower but less than 1000 horsepower are subject to specific emission standards. The specific applicable standards and compliance date of any particular engine or any particular replacement of that engine at any particular point in time is variable, dependent on multiple factors, and can potentially be subject to further changes at the discretion of the owner or operator.

For specific rule requirements, see:

- Section I.D.6.a Applicability and exemptions
- Section I.D.6.b Emission Standards and Compliance deadlines



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Section I.D.6.c Performance Testing

Section I.D.6.d Monitoring and Combustion Process Adjustment

Section I.D.6.e Recordkeeping

Section I.D.6.f Reporting

5.7. Control of Emissions from Stationary and Portable Combustion Equipment in the 8-hour Ozone Control Area or Northern Weld County: Regulation Number 26, Part B, Section II

Replacements stationary internal combustion engines that were located at a major sources of NO_x prior to specific dates must complete inspections and adjustments.

For specific rule requirements, see:

Section II.A.1 Applicability

Section II.A.2 Exemptions

Section II.A.6.b.(iv) & (vii) Combustion Process Adjustment for Engines

Section II.A.6.b.(viii) Compliance dates and frequency

Section II.A.6.c Alternatives

Section II.A.7 Recordkeeping

5.8. Control of Emissions from Specific Major Sources of VOC and/or NO_x in the 8-hour Ozone Control Area: Regulation Number 26, Part B, Section III.

Replacements of units specifically identified in Section III are subject to the additional requirements of the original unit, as specified in Section III.

5.9. NSPS for stationary spark ignition internal combustion engines: 40 CFR Part 60, Subpart JJJJ



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A permanent replacement engine that is manufactured on or after 1/1/09 for emergency engines greater than 25 hp, 7/1/2008 for engines less than 500 hp, 7/1/2007 for engines greater than or equal to 500 hp except for lean burn engines greater than or equal to 500 hp and less than 1,350 hp, and 1/1/2008 for lean burn engines greater than or equal to 500 hp and less than 1,350 hp are subject to the requirements of 40 CFR Part 60, Subpart JJJJ.

Note that the initial test required by NSPS Subpart JJJJ can serve as the testing required by this AOS under Item 6, if approved in advance by the Division, provided that such test is conducted within the time frame specified in Item 6.

Under the provisions of Regulation Number 6 Part B, Section I.C, which are triggered upon the adoption of NSPS JJJJ, relocation of an internal combustion engine (ICE) is defined as the earlier of the date the owner or operator submits an APEN to the Division, or the date the ICE is placed in a secured location where it is intended to be operated within the State of Colorado. However, as of March 23, 2023 the Division has not yet adopted NSPS JJJJ. Until such time as it does, any engine subject to NSPS JJJJ will be subject only under Federal law. Once the Division adopts NSPS JJJJ, determination of the applicable State-Only Enforceable requirements of NSPS JJJJ for the purposes of Regulation Number 6 must be based on the relocation date (if any) as defined in § I.C.1.a. Note that special provisions apply to engines < 500 hp. Refer to PS-Memo 10-03 for additional information on Regulation Number 6 applicability for new and relocated engines.

5.10. Reciprocating internal combustion engine (RICE) MACT: 40 CFR Part 63, Subpart ZZZZ

A permanent replacement engine located at either an area or major source is subject to the requirements in 40 CFR Part 63, Subpart ZZZZ.

Note that the initial test required by the MACT can serve as the testing required by this AOS under Item 6, if approved in advance by the Division, provided that such test is conducted within the time frame specified in Item 6.

6. Portable Analyzer Monitoring



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Note: In some cases there may be conflicting and/or duplicative testing requirements due to overlapping Applicable Requirements. In those instances, please contact the Division's Compliance and Enforcement Unit to discuss streamlining the testing requirements.

Note that the testing required by this Condition may be used to satisfy the portable analyzer monitoring requirements specified by the permit for the relevant time period (i.e. if the permit requires quarterly portable analyzer monitoring, this monitoring event conducted under the AOS will serve as the quarterly monitoring event required by the permit and an additional portable analyzer monitoring event is not required for another three months).

The owner or operator may conduct a reference method test, in lieu of the portable analyzer test required by this Condition, if approved in advance by the Division.

The owner or operator must measure nitrogen oxide (NO_x) and carbon monoxide (CO) emissions in the exhaust from the replacement engine using a portable flue gas analyzer *within seven (7) calendar days of commencing operation of the replacement engine*.

All portable analyzer monitoring required by this permit must be conducted using the Division's Portable Analyzer Monitoring Protocol (ver March 2006 or newer) as found on the Division's web site at: <https://cdphe.colorado.gov/compliance-and-enforcement>

Results of the portable analyzer monitoring must be used to monitor the compliance status of this unit, except for any specifically applicable emission limitations or standards for which an applicable requirement specifies a performance test or Continuous Emission Monitoring System as the compliance monitoring method.

For comparison with an annual (tons/year) or short term (lbs/unit of time) emission limit, the results of the monitoring event must be converted to a lb/hr basis and multiplied by the allowable operating hours in the month or year (whichever applies) in order to monitor compliance. If a source is not limited in its hours of operation the test results will be multiplied by the maximum number of hours in the month (744) or year (8760), whichever applies.

For comparison with a short-term limit that is either input based (lb/mmBtu), output based (g/hp-hr) or concentration based (ppmvd @ 15% O₂) that the existing unit is currently subject to



or the replacement engine will be subject to, the results of the test must be converted to the appropriate units as described in the above-mentioned Portable Analyzer Monitoring Protocol document.

If the portable analyzer results indicate compliance with both the NO_x and CO emission limitations, in the absence of credible evidence to the contrary, the owner or operator may certify that the engine is in compliance with both the NO_x and CO emission limitations for the relevant time period.

Subject to the provisions of C.R.S. 25-7-123.1 and in the absence of credible evidence to the contrary, if the portable analyzer results fail to demonstrate compliance with either the NO_x or CO emission limitations, the engine will be considered to be out of compliance from the date of the portable analyzer test until a portable analyzer test indicates compliance with both the NO_x and CO emission limitations or until the engine is taken offline.